

Bibliography & Background Report

EEG-Based Brain–Computer Interface
Controlling a Robotic Arm

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Prepared as the literature and background report for the case-study project. The report surveys the scientific foundations of electroencephalography-based brain–computer interfaces and their application to the non-invasive control of assistive robotic arms, moving from the general field toward the specific pipeline adopted in this project.

Abstract

Millions of people live with paralysis or severe motor impairment following stroke, spinal-cord injury, or neurodegenerative disease. In many of these conditions the motor cortex continues to generate the intention to move even though the neural signal can no longer reach the muscles. A brain–computer interface (BCI) exploits exactly this residual activity: it records neural signals, decodes the user’s intent, and translates that intent into commands for an external device. This report reviews the scientific literature that underpins a case-study project in which a non-invasive, electroencephalography (EEG) based BCI is used to control a multi-degree-of-freedom robotic arm through motor imagery. Beginning with the foundational definition of BCIs and the neurophysiology of sensorimotor rhythms — in particular event-related desynchronisation and synchronisation (ERD/ERS) — the review narrows toward the concrete signal-processing pipeline used here: EEG acquisition (including the standard BCI Competition IV 2a benchmark), band-pass filtering and independent component analysis for artefact removal, feature extraction based on ERD/ERS, power spectral density and the Common Spatial Pattern algorithm, and classification with Support Vector Machines, Logistic Regression, and compact convolutional networks such as EEGNet. The decoded classes are mapped to seven robotic-arm movement commands enabling reach, grasp, and transport of everyday objects. The report also situates the project against landmark demonstrations, from invasive intracortical control of robotic limbs by people with tetraplegia to recent non-invasive finger-level decoding. The literature confirms that an EEG-BCI robotic-arm pipeline is a viable, modular, and scalable route toward restoring practical independence, while highlighting the open challenges of accuracy, robustness, and real-time embedded deployment that motivate the present work.

Keywords: brain–computer interface; electroencephalography; motor imagery; event-related desynchronisation; common spatial patterns; assistive robotics; robotic arm control; neuroprosthetics.

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1. Introduction and Background

The ambition to control machines directly with thought is no longer confined to science fiction. Over the past three decades, brain–computer interfaces have matured from laboratory curiosities into a recognised branch of neural engineering with concrete clinical relevance. In the canonical definition offered by Wolpaw and colleagues, a BCI is “a communication system in which messages or commands that an individual sends to the external world do not pass through the brain's normal output pathways of peripheral nerves and muscles” [1]. Instead, the system measures brain activity directly, extracts informative features, and converts them into control signals. This bypassing of the neuromuscular route is precisely what makes BCIs so valuable for people whose peripheral pathways are damaged.

The motivating clinical picture for this project is paralysis. Following stroke, traumatic spinal-cord injury, or motor-neuron disease, the descending pathways that carry movement commands from the cortex to the limbs can be interrupted while the cortex itself remains functional. Population estimates place roughly 5.4 million people in the United States alone living with some form of paralysis [16], and stroke — the leading acquired cause of adult disability — accounts for well over ten million new cases worldwide every year [17]. For these individuals the intention to reach for a cup, to grasp it, and to bring it to the mouth is cortically intact yet cannot be executed. A BCI that reads that intention and drives a robotic arm therefore addresses a genuine and large unmet need.

This report is organised to move deliberately from the general to the specific. It first establishes what BCIs are and the neurophysiological phenomena that make non-invasive control possible. It then reviews the clinical demonstrations that prove neural signals can drive robotic limbs. From there it narrows onto the exact processing pipeline adopted in the project — acquisition, preprocessing, feature extraction, classification, and command mapping — grounding each stage in the relevant literature. Finally it surveys the current state of the art and draws conclusions about the viability and future of the approach.

2. Foundations of Brain–Computer Interfaces

2.1 What a BCI is and why EEG is used

Wolpaw et al.'s 2002 review remains the reference point for the field; it framed BCIs as a fundamentally new output channel for the brain and catalogued the signal types, translation algorithms, and applications that define the discipline [1]. A central distinction it drew is between invasive approaches, which place electrodes on or within cortical tissue, and non-invasive approaches, which record from the scalp. Electroencephalography — the measurement of the summed post-synaptic potentials of cortical neurons at the scalp — has become the dominant non-invasive modality because it is inexpensive, portable, safe, and offers millisecond temporal resolution. These properties make EEG the natural choice for an assistive system intended to operate outside a clinical laboratory, and it is the modality adopted in this project.

2.2 Sensorimotor rhythms and motor imagery

The physiological signal that makes EEG-based movement control possible is the modulation of sensorimotor rhythms. Over the sensorimotor cortex the EEG contains characteristic oscillations — the mu rhythm (roughly 8–13 Hz) and the beta rhythm (roughly 13–30 Hz). Pfurtscheller and Lopes da Silva showed that the amplitude of these rhythms falls in the moment before and during movement, a phenomenon termed event-related desynchronisation (ERD), and rebounds afterwards as event-related synchronisation (ERS) [2]. Crucially, the same ERD/ERS pattern appears not only during actual movement but also during motor imagery — the mere mental rehearsal of a movement without any muscle activation [3]. Because imagined left-hand, right-hand, foot, or tongue movements each produce a spatially distinct

ERD/ERS signature over the cortex, these patterns can serve as reliable, voluntarily controllable features for a classifier. Motor imagery is therefore the control paradigm at the heart of this project: the user imagines a movement, the system detects the associated spectral-spatial change, and it issues the corresponding robotic command.

3. From Neural Intent to Robotic Action

If sensorimotor rhythms provide the readable signal, the clinical proof that neural intent can drive a physical limb comes from landmark demonstrations in people with paralysis. The most influential is the BrainGate study by Hochberg and colleagues, published in *Nature* in 2012, in which two people with long-standing tetraplegia used an intracortical BCI to control a robotic arm for three-dimensional reach-and-grasp tasks [4]. One participant, paralysed for nearly fifteen years, used the arm to lift a bottle, bring it to her mouth, and drink unaided. Although that work used invasive microelectrode arrays rather than scalp EEG, it established the essential principle on which this project rests: the motor cortex retains the capacity to specify movement long after the body can execute it, and a decoding system can recover that specification and act on it.

The scientific question this project addresses is whether a comparable capability can be achieved non-invasively. A scalp-EEG system trades the fine spatial resolution and high information rate of implanted electrodes for safety, low cost, and immediate accessibility — a trade-off that is attractive for widely deployable assistive technology. Demonstrating that motor imagery decoded from EEG can command a multi-degree-of-freedom arm to perform useful daily tasks is precisely the gap the project sets out to fill.

4. Materials and Methods

This section reviews, stage by stage, the literature supporting the signal-processing pipeline used in the project. The pipeline follows the classical BCI structure — acquisition, preprocessing, feature extraction, classification, and command execution — each stage anchored in established methods.

4.1 EEG data acquisition and datasets

Signals may be obtained either from a physical EEG headset worn during motor-imagery tasks or from a public benchmark dataset that allows algorithms to be developed and compared reproducibly. The de facto standard benchmark is the BCI Competition IV dataset 2a, released by Graz University of Technology and reviewed by Tangermann et al. [9]. It comprises recordings from nine healthy participants performing four motor-imagery classes — left hand, right hand, both feet, and tongue — acquired from 22 EEG channels arranged according to the international 10–20 system, sampled at 250 Hz, with two sessions per subject and 288 trials per session. Its balanced, well-documented structure makes it the natural choice for developing and validating the classifiers used here before migrating to live hardware.

4.2 Signal preprocessing and artefact removal

Raw EEG is contaminated by noise and by non-cerebral artefacts, most notably ocular activity from blinks and eye movements and electromyographic activity from facial and neck muscles. Two steps address this. First, a band-pass filter restricts the signal to the 8–30 Hz range that spans the mu and beta rhythms carrying motor-imagery information, suppressing slow drifts and high-frequency muscle noise. Second, Independent Component Analysis (ICA) decomposes the multi-channel recording into maximally independent sources; components corresponding to eye and muscle artefacts are identified and removed before the signal is reconstructed. The foundational treatment of this technique by Jung, Makeig and

colleagues showed that blind source separation can isolate and subtract such artefacts without discarding contaminated data segments, preserving the neural signal of interest [15]. Together these steps yield the clean, band-limited signal that the feature-extraction stage requires.

4.3 Feature extraction

The goal of feature extraction is to summarise each trial as a compact vector that maximally separates the motor-imagery classes. Three complementary approaches inform the project. The most direct exploits the ERD/ERS phenomenon itself, quantifying the band-power decrease over the contralateral sensorimotor cortex during imagery [2, 3]. A related measure is the power spectral density computed per trial window, which captures how energy is distributed across the mu and beta bands. The most powerful spatial method is the Common Spatial Pattern (CSP) algorithm introduced for single-trial EEG by Ramoser et al. [5]: CSP learns spatial filters that maximise the variance of the signal for one class while minimising it for the other, producing highly discriminative features. Blankertz and colleagues later provided a practical guide to making CSP robust against noise and non-stationarity [6], and Ang et al. extended it to the Filter Bank Common Spatial Pattern, which optimises the frequency band jointly with the spatial filter and has become a standard high-performing baseline [7]. Xygonakis et al. further demonstrated that applying CSP in the EEG source space, after estimating the cortical origin of the scalp signals, can improve multi-class motor-imagery decoding [10].

4.4 Classification and command mapping

Once features are extracted, a classifier assigns each trial to a motor-imagery class. Support Vector Machines are the workhorse of the field: paired with CSP features they provide strong, well-regularised decoding of motor imagery and motor execution, and this CSP-plus-SVM combination is the primary decoder adopted in the project. Logistic Regression offers a lightweight, interpretable alternative suited to embedded deployment. More recently, deep learning has been applied end-to-end: Lawhern et al.'s EEGNet, a compact convolutional neural network using depthwise and separable convolutions, learns spatial and temporal features directly from raw EEG and generalises across several BCI paradigms while remaining small enough for real-time use [8]. Whichever decoder is chosen, its output is mapped to one of the discrete robotic commands, translating an abstract class label into a concrete joint movement.

4.5 Robotic-arm control

The final stage converts the classifier output into physical action. The project defines seven essential movement commands — base rotation, forward reach, backward retraction, elbow flexion, wrist rotation, gripper open, and gripper close — which together span the degrees of freedom needed for a full repertoire of everyday assistive tasks such as reaching for, grasping, and transporting objects. This design mirrors the reach-and-grasp objectives validated in the neural-control literature [4] and keeps the mapping between imagined intent and end-effector motion as intuitive as possible, a factor repeatedly identified as decisive for usable BCI control.

5. Related Work and State of the Art

Placing the project against current research clarifies both its feasibility and its ambition. Meng et al. showed that healthy users can continuously control a robotic arm for reach-and-grasp tasks using non-invasive EEG alone, demonstrating that scalp signals — not only implanted electrodes — can drive multi-dimensional robotic movement [11]. Subsequent work has pushed decoding resolution and practicality further: recent studies have optimised motor-imagery paradigms specifically for robotic-arm control [12], and a 2025 report in *Nature Communications* demonstrated real-time non-invasive control of a robotic hand at the level of individual fingers, achieving around 80% accuracy for two-finger imagery

tasks [14]. Hybrid interfaces are another active direction; a system combining EEG with electrooculography, for example, was used to operate an integrated wheelchair-and-robotic-arm platform, improving reliability by fusing complementary signals [13]. Collectively these works confirm that the project's architecture sits on a well-supported trajectory, while the persistent themes of limited accuracy, inter-session variability, and the need for intuitive mappings define the challenges the project must navigate.

6. Conclusion

The literature reviewed here supports a clear and coherent conclusion. Brain–computer interfaces rest on a firm neurophysiological foundation: sensorimotor rhythms modulate in a spatially specific way during motor imagery, and this modulation can be measured non-invasively with EEG and decoded into discrete intentions [1, 2, 3]. Clinical demonstrations have already proven that decoded neural intent can drive a robotic arm to perform meaningful reach-and-grasp tasks for people with paralysis [4], and a growing body of work shows that comparable control is achievable non-invasively [11, 14].

Building on these foundations, the project assembles a complete, end-to-end pipeline — EEG acquisition on a standard benchmark [9], band-pass filtering and ICA-based artefact removal [15], feature extraction via ERD/ERS, power spectral density and the Common Spatial Pattern family [5, 6, 7, 10], classification with Support Vector Machines, Logistic Regression, or EEGNet [8], and mapping to seven robotic-arm commands. Each stage is grounded in established, independently validated methods, and each is modular and independently upgradeable, so that better classifiers or arms with more degrees of freedom can be substituted without redesigning the whole system.

The central message is that an EEG-based BCI for robotic-arm control is not a speculative proposition but a viable, non-invasive, and scalable pathway from neural intent to physical action. It directly addresses an unmet need affecting millions of people living with paralysis. The principal open challenges — improving single-trial accuracy, ensuring robustness across sessions and users, and achieving reliable real-time operation on embedded hardware — are precisely the problems that motivate the continued development of the system described in this report. Future work will focus on real-time EEG hardware integration, closed-loop haptic feedback, and an expanded motor-imagery vocabulary to enrich the range of controllable actions.

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